

Question number	Answer	Additional guidance	Mark
8(a)(i)	A description that includes the following points: - gene is the {length of DNA / sequence of bases} coding for a (poly)peptide (1) - allele is the (different) {version / form / variation} of the gene (1) - gene for feather colour <u>and</u> allele for {white / black} feathers (1)	ACCEPT protein / sequence of amino acids NB piece together DO NOT ACCEPT speckled feathers	(3)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	A description that includes the following points: - genotype is the combination of alleles (1) - phenotype is the {(observable) characteristics / appearance / traits} (1) - phenotype is the colour of the feathers <u>and</u> genotype is the presence of white and or black alleles (1)	ACCEPT mixture of alleles / the alleles present DO NOT ACCEPT genes ACCEPT feature / ref to colour of feathers IGNORE ref to environment ACCEPT phenotype is {white / black / speckled / mixture of black and white} (feathers) genotype is {WW / BB / WB} or any other letters used NB piece together	(3)

Question number	Answer	Additional guidance	Mark
8(b)	An answer that includes the following points: - parent's genotypes shown as BB and BW (1) - offspring's genotypes shown as BB and BW (1) - number of speckled chicks given as 12 or 13 (1)	ACCEPT other letters, including B and b DO NOT ACCEPT X and Y B for white and W for black ACCEPT other letters, including B and b CE from 1 CE from 2 provided whole number NB Bald answer = 1 mark	(3)

Question number	Answer	Additional guidance	Mark
8(c)(i)	Chi squared (1)	ACCEPT Chi square / X squared / x squared / X^2 / x^2 / χ^2 distribution (test) / chi (test) / closeness of fit / goodness of fit phonetic spellings e.g kai / cai / khi	(1)

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8(c)(ii)	An answer that includes the following points: <table border="1"> <thead> <tr> <th rowspan="2">Steps in the calculation for the statistics test</th><th colspan="3">Colour of feathers of chicks</th></tr> <tr> <th>Speckled</th><th>White</th><th>Black</th></tr> </thead> <tbody> <tr> <td>Observed number (O)</td><td>243</td><td>125</td><td>112</td></tr> <tr> <td>Expected number (E)</td><td>240</td><td>120</td><td>120</td></tr> <tr> <td>(O-E)</td><td>3</td><td>5</td><td>-8</td></tr> <tr> <td>$\frac{(O-E)^2}{E}$</td><td>0.0375 9/240 3/80</td><td>0.2083 25/120 5/24</td><td>0.5333 64/120 8/15</td></tr> </tbody> </table>	Steps in the calculation for the statistics test	Colour of feathers of chicks			Speckled	White	Black	Observed number (O)	243	125	112	Expected number (E)	240	120	120	(O-E)	3	5	-8	$\frac{(O-E)^2}{E}$	0.0375 9/240 3/80	0.2083 25/120 5/24	0.5333 64/120 8/15	All correct = 2 marks One row correct = 1 mark ACCEPT 8 DO NOT ACCEPT -3 / -5 ACCEPT 2 or 3 or 4 dps but not necessarily consistently the same number. DO NOT ACCEPT recurring	(2)
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8(c)(iii)	0.8 / 0.78 / 0.779 / 0.7791 (1)	CE / rounding effect from (c)(ii) DO NOT ACCEPT 0 NB 187/240 = 0.7792	(1)

Question number	Answer	Additional guidance	Mark
8(c)(iv)	An explanation that includes two of the following points: • {calculate / use} the number of degrees of freedom • use (a probability) value of (up to or equal to) 5%/0.05 (1) • compare (calculated) {value / result} to (critical) value (1) • if the (calculated) {value / result} is greater than (critical) value then null hypothesis is rejected (1)	ACCEPT description of how to work out degrees of freedom NB mp 4 alone = 2 marks ACCEPT the converse argument any quoted figures for the calculated value	(2)